

Lecture 14: Backward Stability; Vector and Matrix Norms

Math 6610

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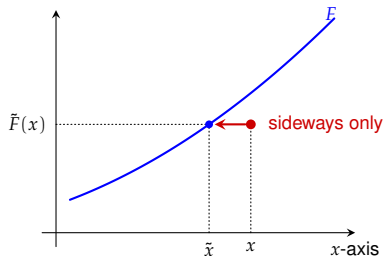
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Today's Goals

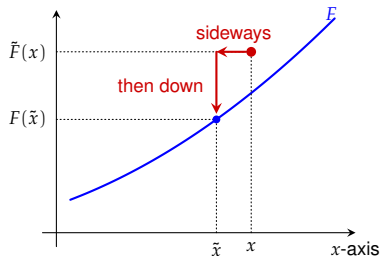
- Prove the accuracy theorem $\|\tilde{y} - y\| / \|y\| = O(\kappa \epsilon_{\text{mach}})$
- Review vector norms $(1, 2, \infty)$ and the Frobenius norm
- Define induced (lub) matrix norms and compute the classical three
- Prove submultiplicativity $\text{lub}(AB) \leq \text{lub}(A) \text{lub}(B)$

Stable vs. Backward Stable

Backward stable



Stable



sideways: $|\tilde{x} - x| \leq C\epsilon_{\text{mach}} |x|$,

down: $|\tilde{F}(x) - F(\tilde{x})| \leq C\epsilon_{\text{mach}} |F(\tilde{x})|$

Accuracy Theorem

Remark (Backward stable (Lecture 13)). The computed answer is the exact answer for a slightly changed input: for each $x \neq 0$ there is $\tilde{x}$ with

$$\tilde{F}(x) = F(\tilde{x}), \quad \|\tilde{x} - x\| \leq C \epsilon_{\text{mach}} \|x\| \quad \text{for all small } \epsilon_{\text{mach}},$$

with C the same for every x .

Definition (Condition number, recalled from Lecture 9). For $\delta > 0$ let $g(\delta)$ be the worst amplification over all input changes δx with $0 < \|\delta x\| \leq \delta$:

$$g(\delta) := \sup_{0 < \|\delta x\| \leq \delta} \frac{\|F(x + \delta x) - F(x)\| / \|F(x)\|}{\|\delta x\| / \|x\|}, \quad \kappa(x) := \limsup_{\delta \rightarrow 0} g(\delta).$$

(“sup” = the largest value, or the limit of the values if no largest one exists.)

Accuracy of a backward stable algorithm

Theorem (Accuracy of backward stable algorithms). Let $\tilde{F} : X \rightarrow Y$ be a backward stable algorithm for $F : X \rightarrow Y$. Let $x \in X$ with $x \neq 0$, $F(x) \neq 0$, and $0 < \kappa(x) < \infty$. Then there is $\epsilon_0 > 0$ (depending on x) such that

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} \leq 2C \kappa(x) \epsilon_{\text{mach}} \quad \text{whenever } \epsilon_{\text{mach}} \leq \epsilon_0,$$

with C from backward stability. In short:

$$\|\tilde{F}(x) - F(x)\| / \|F(x)\| = O(\kappa(x) \epsilon_{\text{mach}}).$$

Accuracy Theorem: Some Intuitions

Remark.

- 1 The algorithm hands back $F(\tilde{x})$: the exact answer for an input that is wrong by a relative amount of at most $C \epsilon_{\text{mach}}$ (backward stability).
- 2 The problem turns a small relative input error into a relative output error at most about $\kappa(x)$ times larger (conditioning).

So

$$\underbrace{\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|}}_{\text{output error}} = \underbrace{\frac{\text{output error}}{\text{input error}}}_{\lesssim \kappa(x)} \times \underbrace{\frac{\|\tilde{x} - x\|}{\|x\|}}_{\text{input error} \leq C \epsilon_{\text{mach}}} \lesssim \kappa(x) \cdot C \epsilon_{\text{mach}}.$$

Nothing else about the algorithm is used: every backward stable algorithm gets this bound for free.

Remark. “About $\kappa(x)$ ” holds only for small input changes. The proof makes this precise: for small ϵ_{mach} , “about $\kappa(x)$ ” becomes “at most $2\kappa(x)$ ”.

Accuracy Theorem: Proof

Proof. Idea: error = (amplification) $\times$ (input error) $\leq 2\kappa(x) \cdot C \epsilon_{\text{mach}}$.

1. *Error as an input change.* Backward stability gives $\tilde{x} = x + \delta x$ with

$$\tilde{F}(x) = F(x + \delta x), \quad \frac{\|\delta x\|}{\|x\|} \leq C \epsilon_{\text{mach}}.$$

(If $\delta x = 0$ the error is 0; so let $\delta x \neq 0$.)

2. *Split into two factors.* Multiply and divide by $\|\delta x\| / \|x\|$:

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} = \underbrace{\frac{\|F(x + \delta x) - F(x)\| / \|F(x)\|}{\|\delta x\| / \|x\|}}_{\text{amplification}} \cdot \underbrace{\frac{\|\delta x\|}{\|x\|}}_{\leq C \epsilon_{\text{mach}}}.$$

(continued on the next slide)

Accuracy Theorem: Proof (Cont'd)

Proof (continued). 3. Choose how small is “small”. As $\delta \rightarrow 0$, $g(\delta)$ comes down to $\kappa(x)$. Since $2\kappa(x) > \kappa(x)$, it is eventually below $2\kappa(x)$: there is $\delta_0 > 0$ with

$$g(\delta_0) \leq 2\kappa(x).$$

4. Our change is that small. If $\epsilon_{\text{mach}} \leq \epsilon_0 := \delta_0 / (C \|x\|)$, then by Step 1

$$\|\delta x\| \leq C \epsilon_{\text{mach}} \|x\| \leq \delta_0.$$

5. Bound the amplification. Our δx is one of the changes counted in $g(\delta_0)$, so

$$\text{amplification of } \delta x \leq g(\delta_0) \leq 2\kappa(x).$$

Put this into Step 2:

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} \leq 2\kappa(x) \cdot C \epsilon_{\text{mach}}.$$



Vector Norms

Definition (Norm). A norm on $\mathbb{C}^n$ is a function $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}$ (a “length”) such that for all $x, y \in \mathbb{C}^n$, $\alpha \in \mathbb{C}$:

- ❶ $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$
- ❷ $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)
- ❸ $\|\alpha x\| = |\alpha| \|x\|$

Example (The 1-, 2- and ∞ -norms).

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

(Total size, Euclidean length, largest entry.)

Matrix Norms

Definition (Matrix norm). A matrix norm is a function $\|\cdot\| : \mathbb{C}^{n \times n} \rightarrow \mathbb{R}$ such that for all $A, B \in \mathbb{C}^{n \times n}$, $\alpha \in \mathbb{C}$:

- 1 $\|A\| \geq 0$, and $\|A\| = 0$ if and only if $A = 0$
- 2 $\|A + B\| \leq \|A\| + \|B\|$
- 3 $\|\alpha A\| = |\alpha| \|A\|$

Example (Vec norm). Stack the columns of A into one long vector,

$$\text{vec}(A) := [a_{11} \ a_{21} \ \cdots \ a_{n1} \ a_{12} \ \cdots \ a_{nn}]^T \in \mathbb{C}^{n^2},$$

and set $\|A\| := \|\text{vec}(A)\|$ for any vector norm on $\mathbb{C}^{n^2}$.

Example (Frobenius norm).

$$\|A\|_F := \|\text{vec}(A)\|_2 = \sqrt{\sum_{j,k=1}^n |a_{jk}|^2}.$$

Induced (lub) Matrix Norms

Definition (Induced matrix norm). Let $\|\cdot\|$ be a vector norm on $\mathbb{C}^n$. For $A \in \mathbb{C}^{n \times n}$,

$$\text{lub}(A) = \text{lub}_{\|\cdot\|}(A) := \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\| :$$

the largest factor by which A stretches a vector (lub = least upper bound).

Remark (Basic inequality). No vector is stretched more than the largest stretch: $\|Ax\| \leq \text{lub}(A) \|x\|$ for all $A \in \mathbb{C}^{n \times n}$, $x \in \mathbb{C}^n$ (for $x = 0$ both sides are 0).

lub Is a Matrix Norm

Proposition. For every vector norm $\|\cdot\|$ on $\mathbb{C}^n$, $\text{lub}_{\|\cdot\|}$ is a matrix norm, and $\text{lub}(I) = 1$.

Proof. In Steps 2–4, fix x with $\|x\| = 1$, then take the maximum over all such x .

1. *Positive.* $\text{lub}(A) \geq 0$, as a maximum of norms. If $\text{lub}(A) = 0$, then $\|Ax\| \leq \text{lub}(A) \|x\| = 0$ for every x , so

$$Ae_k = 0 \quad (k = 1, \dots, n) \implies \text{every column of } A \text{ is } 0 \implies A = 0.$$

2. *Triangle inequality.* Vector triangle inequality, then the definition of lub :

$$\|(A + B)x\| \leq \|Ax\| + \|Bx\| \leq \text{lub}(A) + \text{lub}(B).$$

3. *Scaling.* Vector scaling rule, for every x :

$$\|(\alpha A)x\| = |\alpha| \|Ax\| \implies \text{lub}(\alpha A) = |\alpha| \text{lub}(A).$$

4. *Identity.* $\|Ix\| = \|x\| = 1$ for every such x , so $\text{lub}(I) = 1$. ■

The Three Classical lub Norms

Proposition. For $A \in \mathbb{C}^{n \times n}$:

Norm on $\mathbb{C}^n$	Induced matrix norm
$\ \cdot\ _1$	$\ A\ _1 := \text{lub}_{\ \cdot\ _1}(A) = \max_{1 \leq k \leq n} \sum_{j=1}^n a_{jk} $ (max column sum)
$\ \cdot\ _\infty$	$\ A\ _\infty := \text{lub}_{\ \cdot\ _\infty}(A) = \max_{1 \leq j \leq n} \sum_{k=1}^n a_{jk} $ (max row sum)
$\ \cdot\ _2$	$\ A\ _2 := \text{lub}_{\ \cdot\ _2}(A) = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^H A)}$

Remark (Words). A^H = conjugate transpose. $\lambda_{\max}(A^H A)$ = largest eigenvalue of $A^H A$ (it is real and ≥ 0). $\sigma_{\max}(A)$ = largest singular value of A .

Submultiplicativity

Theorem (*lub norms are submultiplicative*).

$$\text{lub}(AB) \leq \text{lub}(A) \text{lub}(B) \quad \text{for all } A, B \in \mathbb{C}^{n \times n}.$$

Proof. Idea: $ABx = A(Bx)$: first B stretches, then A ; the factors multiply.

1. *One vector.* Let $\|x\| = 1$. Apply $\|My\| \leq \text{lub}(M) \|y\|$ with $M = A$, $y = Bx$, then with $M = B$, $y = x$:

$$\|ABx\| \leq \text{lub}(A) \|Bx\| \leq \text{lub}(A) \text{lub}(B) \|x\| = \text{lub}(A) \text{lub}(B).$$

2. *All vectors.* Maximize over $\|x\| = 1$: the left side becomes $\text{lub}(AB)$. ■

Remark (Powers). $\|A^k\| \leq \|A\|^k$ for $k = 1, 2, \dots$: induction on k , base case $k = 1$ trivial, and

$$\|A^{k+1}\| = \|AA^k\| \leq \|A\| \|A^k\| \leq \|A\| \|A\|^k = \|A\|^{k+1}.$$

The 1-Norm Formula

Proposition. Let $a_1, \dots, a_n$ be the columns of A . Then

$$\|A\|_1 = \max_k \|a_k\|_1 \quad (\text{max column sum}).$$

Remark.

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n, \quad \|x\|_1 = |x_1| + \cdots + |x_n| = 1.$$

The weights have total size 1, so the mix is never bigger than the biggest column; all weight on the biggest column gives exactly that column.

Example ($A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$). Column sums: $|1| + |3| = 4$, $|-2| + |4| = 6$, so $\|A\|_1 = 6$.

$$x = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} : Ax = \begin{bmatrix} -1/2 \\ 7/2 \end{bmatrix}, \|Ax\|_1 = 4 \leq 6; \quad x = e_2 : Ax = \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \|Ax\|_1 = 6.$$

The 1-Norm Formula: Proof

Proof. Let $M := \max_k \|a_k\|_1$. We show: every x with $\|x\|_1 = 1$ gives $\|Ax\|_1 \leq M$, and one x gives exactly M .

1. *Upper bound.* Let $\|x\|_1 = 1$. Write Ax as a mix of columns, use the triangle inequality, then $\|a_k\|_1 \leq M$:

$$\begin{aligned}\|Ax\|_1 &= \|x_1 a_1 + \cdots + x_n a_n\|_1 \leq |x_1| \|a_1\|_1 + \cdots + |x_n| \|a_n\|_1 \\ &\leq M(|x_1| + \cdots + |x_n|) = M \|x\|_1 = M.\end{aligned}$$

2. *The bound is reached.* Let column k^* have $\|a_{k^*}\|_1 = M$. Put all weight on it, $x = e_{k^*}$:

$$\|e_{k^*}\|_1 = 1, \quad Ae_{k^*} = a_{k^*}, \quad \|Ae_{k^*}\|_1 = M.$$

3. *Conclusion.* The largest stretch is M : $\|A\|_1 = M$. ■

Remark (Cost). $\|A\|_1$ and $\|A\|_\infty$ take about n^2 operations: cheap.

The ∞ -Norm Formula

Proposition.

$$\|A\|_{\infty} = \max_j \sum_k |a_{jk}| \quad (\text{max row sum}).$$

Remark (*Key fact: each entry of Ax is a row times x .*)

$$(Ax)_j = a_{j1}x_1 + \cdots + a_{jn}x_n, \quad \|x\|_{\infty} = 1 \iff \text{all } |x_k| \leq 1, \text{ at least one } = 1.$$

Entry j is largest when no terms cancel: every $a_{jk}x_k$ points the same way.

Example ($A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$). Row sums: $|1| + |-2| = 3$, $|3| + |4| = 7$, so $\|A\|_{\infty} = 7$.

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix} : Ax = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \|Ax\|_{\infty} = 3 \leq 7; \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} : Ax = \begin{bmatrix} -1 \\ 7 \end{bmatrix}, \|Ax\|_{\infty} = 7.$$

(Second choice: signs of row 2, so $3 \cdot 1 + 4 \cdot 1$ has no cancellation.)

The ∞ -Norm Formula: Think After Class

Remark (Exercise: prove the formula). Let $M := \max_j \sum_k |a_{jk}|$. Mimic the 1-norm proof.

- 1 *Upper bound.* For $\|x\|_\infty = 1$, show $|(Ax)_j| \leq M$ for every row j .
- 2 *Reached.* For a row j^* with sum M , find x with $|x_k| = 1$ and $a_{j^*k}x_k = |a_{j^*k}|$.
(Real A : which sign? Complex A : try $\overline{a_{j^*k}}/|a_{j^*k}|$.)
- 3 *Compare.* Why does $x = e_k$, best for $\|\cdot\|_1$, fail here? Test it on the example.

Remark (Bonus). Show $\|A\|_\infty = \|A^T\|_1$.

The 2-Norm

Remark (No entrywise formula).

$$\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^H A)}$$

needs eigenvalues, not just sums of entries (proved later in the course).

Proposition. If $U \in \mathbb{C}^{n \times n}$ is unitary ($U^H U = I$), then $\|Ux\|_2 = \|x\|_2$ for all $x \in \mathbb{C}^n$, and $\|U\|_2 = 1$.

Proof. Idea: like a rotation or reflection, U keeps lengths.

1. *Lengths are kept.* Use $\|v\|_2^2 = v^H v$, then $U^H U = I$:

$$\|Ux\|_2^2 = (Ux)^H (Ux) = x^H U^H U x = x^H x = \|x\|_2^2.$$

2. *Largest stretch.* Every stretch equals 1:

$$\|U\|_2 = \max_{\|x\|_2=1} \|Ux\|_2 = \max_{\|x\|_2=1} \|x\|_2 = 1.$$

Which Norm When?

Proposition. For all $x \in \mathbb{C}^n$:

$$\|x\|_{\infty} \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_{\infty}.$$

Proof. 1. Largest entry $\leq$ length. Let $|x_m| = \|x\|_{\infty}$. One term $\leq$ the sum; then take square roots:

$$\|x\|_{\infty}^2 = |x_m|^2 \leq \sum_i |x_i|^2 = \|x\|_2^2.$$

2. Length $\leq$ total size. Cross terms are ≥ 0 ; then take square roots:

$$\|x\|_1^2 = \sum_i |x_i|^2 + \sum_{i \neq l} |x_i| |x_l| \geq \|x\|_2^2.$$

3. Total size $\leq n \times$ largest entry. Each of the n terms is at most $\|x\|_{\infty}$, so $\|x\|_1 = \sum_{i=1}^n |x_i| \leq n \|x\|_{\infty}$. ■

Which Norm When? (Cont'd)

Remark (*All norms are comparable; proof omitted*). For any two norms $\|\cdot\|_a$, $\|\cdot\|_b$ on $\mathbb{C}^n$ there are $c, c' > 0$ with $c \|x\|_a \leq \|x\|_b \leq c' \|x\|_a$ for all x . So “ $= O(\epsilon_{\text{mach}})$ ” or “converges” in one norm means the same in every norm.

- **Constants change.** $\kappa_1(A)$, $\kappa_2(A)$, $\kappa_\infty(A)$ differ in general, by factors that depend only on n . Say which norm you use.
- $\|\cdot\|_1$, $\|\cdot\|_\infty$: easy to compute (about n^2 operations); good for bounds and in software.
- $\|\cdot\|_2$: unchanged by unitary factors, $\|UAV\|_2 = \|A\|_2$ for unitary U, V (rotations and reflections do not change it). The natural norm for eigenvalues, later in the course.
- $\|\cdot\|_F$: also unchanged by unitary factors and easy to compute, but for $n \geq 2$ not an lub norm of any vector norm: $\|I\|_F = \sqrt{n} \neq 1 = \text{lub}(I)$.

Looking Ahead

- **Next lecture:** Conditioning of $Ax = b$; Stability of LU Factorization